

Importance of Multidisciplinary Design Optimization for Wave-Driven Desalination Systems

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Abstract

This is the abstract. Summarize your problem, approach, and key results concisely. You can include equations and references if needed.

1. Introduction

The global freshwater crisis necessitates innovative solutions for clean water production. Demand for freshwater is anticipated to grow by over 40% by 2050 [1], while droughts, urbanization, and uneven distribution of water resources add to the pressure on our existing freshwater supplies. Seawater reverse osmosis (SWRO) is a proven technology for creating freshwater from seawater, but is quite energy intensive, requiring 2-4 kWh/m³ [2]. This cost associated with this high energy demand make up 25%-40% of the total cost of water [3], and when supplied through fossil fuels, this energy consumption further exacerbates water scarcity challenges [4]. To ensure sustainability, it is therefore crucial to power SWRO efficiently through renewable energy sources.

Various renewable energy sources for powering SWRO, such as wind [5] and solar [6], have been explored. Another intriguing solution is wave energy converters (WECs). SWRO and WECs have natural co-location, with SWRO plants typically situated near the coast while WECs are deployed offshore. Beyond both being marine-based technologies, WECs and SWRO plants share complementary operational characteristics [3]. SWRO requires high pressure seawater, and WECs can provide this directly using a hydraulic style power take-off (PTO) system, forming a wave driven desalination system [7]. This direct drive approach eliminates the need for conversion to an intermediate form of energy, potentially improving overall system efficiency and reducing system complexity. This has the potential to give WECs a significant advantage over other renewable energy sources for powering SWRO desalination.

Despite their potential, WECs face significant challenges limiting their widespread adoption. Notably, their high costs pose a major barrier. Therefore, design optimization is essential to improve economic viability. Multi-disciplinary Design Optimization (MDO) provides a systematic framework to address this challenge by simultaneously considering the designs and interactions of multiple subsystems and optimizing overall performance rather than isolated improvements [8]. MDO approaches integrating PTO design with hydrodynamics and/or controls has been demonstrated to significantly improve the design of electricity producing WECs [9, 10, 11, 12]. Michelén Ströfer et al. (2023) see a 22% increase in electrical power output through PTO and control co-design (CCD) [9], while Grasberger et al. (2024) report a 60% improvement in performance using MDO when compared to traditional design methods [12].

The application of MDO to these problems is effective because of the interactions between the WEC, PTO, and SWRO plant. In 2016 Garcia-Rosa and Ringwood [13] investigated the sensitivity of optimal WEC geometry to control strategy, demonstrating the importance of control co-design. Coe et al. have further advanced our understanding of these interactions through impedance matching studies. In 2021 Coe et al. demonstrated how well-established control techniques, causal feedback control and impedance matching, are quite effective for WEC control, but requires a well designed PTO system that properly shapes the system impedance [14]. In 2025, Coe et al. tackle this challenge and apply impedance matching to PTO design [15]. Although the studies by Coe et al. focus on rack and pinion style WEC PTOs for electricity generation, which do not have any rectifying behavior (unlike hydraulic PTOs like what are used in wave-driven desalination systems), impedance matching principles are important to consider for systems with rectifying behavior too [16].

These MDO and impedance matching studies emphasize the importance of utilizing holistic models that account for wave dynamics, PTO performance, and control strategies in electricity-generating WEC optimization. Despite this, these ideas have yet to be applied to wave-driven desalination systems, leaving a critical gap in research.

Yu and Jenne (2017 and 2018) [17, 18], Brodersen et al. (2022) [19], Suchithra et al. (2022) [20] and Simmons and Van de Ven (2023) [21] have all built models that account for the coupled dynamics of wave-driven desalination systems. However, these studies simply evaluate these systems and do not leverage their models for design optimization.

This paper addresses this gap through the development of a holistic MDO framework for wave-driven desalination system design. A simplified wave-driven desalination schematic is shown in Figure 1, where an oscillating surge

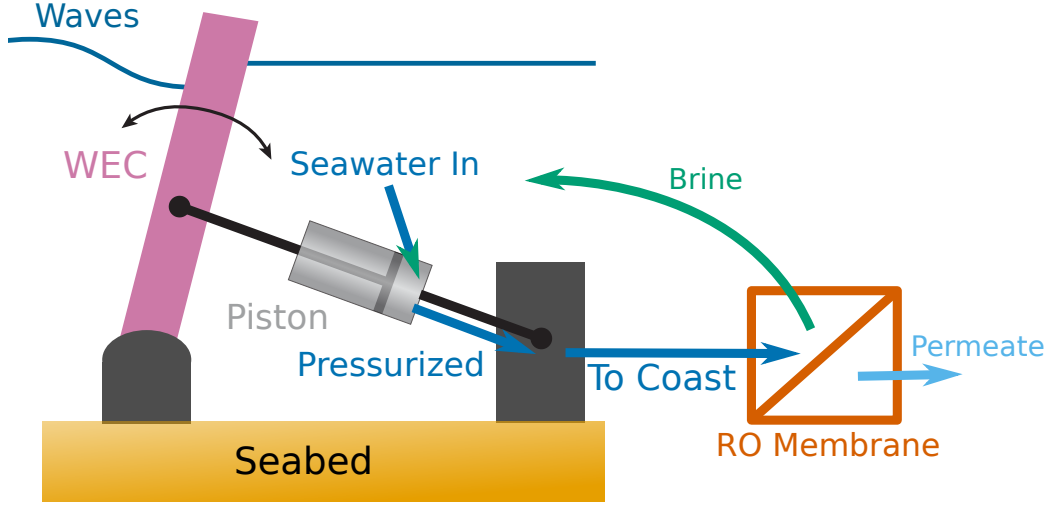


Figure 1: Simple wave-driven desalination system concept sketch.

wave energy converter (OSWEC) drives a hydraulic cylinder, pressurizing seawater and delivering it to an onshore SWRO desalination plant. This system enables freshwater production without the need for major external energy input. While similar systems have been studied in literature [18, 20, 22, 21], this study has a notable difference in that it lacks an energy recovery unit (ERU).

The concept sketch (Figure 1) depicts a high level overview of the wave-driven desalination system but does not capture all the hydraulic components included in this design study. A detailed hydraulic circuit diagram is shown in Figure 2, which includes an accumulator to smooth the pressure delivered to the SWRO plant, a pressure relief valve to avoid operating the SWRO plant beyond its rated capacity, a brine side throttle valve set to control the recovery ratio of the SWRO plant, as well as directional valves.

One of the important design trade-offs to consider for these systems is the size of the accumulator vs the capacity of the SWRO plant. A larger accumulator offers more smoothing of the flow delivered to the SWRO plant. Since the SWRO plant can not operate above a certain capacity, a larger accumulator therefore allows for a smaller SWRO plant to be used without losing water through the pressure relief valve. Since the SWRO plant is a major cost driver for the system [17], investing in larger more expensive accumulators appears to be a massive cost saving strategy. However, as Michelén Ströfer et al. (2023) show, changing the impedance of the PTO system has a significant impact on the performance of a WEC [9]. Therefore, it is important to consider the impact changing the accumulator size has on the system dynamics while making these economic trade-offs.

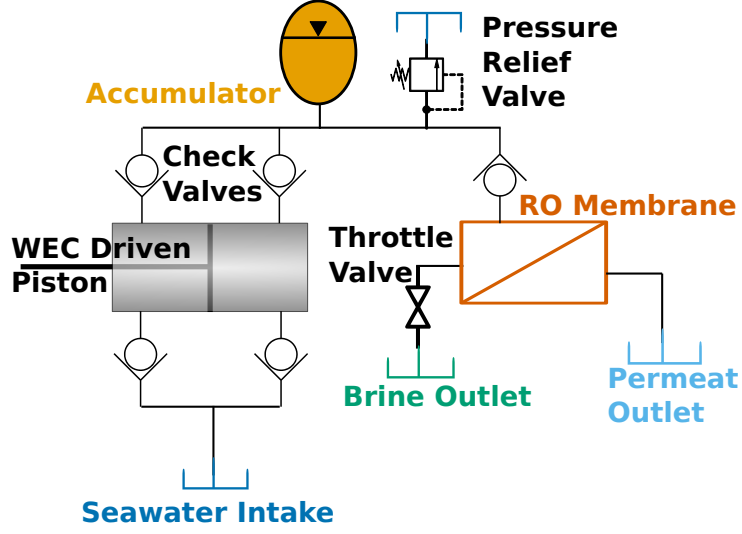


Figure 2: Hydraulic circuit diagram of the wave-driven desalination system.

In addition to the higher usage rate argument, another common argument for large accumulators is that SWRO traditionally operates at steady state conditions. Lai et al. (2016) investigate the effectiveness of various smoothing methods for wind powered SWRO [23]. Hopkins et al. (2014) do a similar study for wave-driven desalination, looking at various accumulator geometries [24]. However, the study by Hopkins et al. do not investigate the impact of accumulator design on WEC performance, ignoring the interactions between the WEC and PTO. Bacelli et al. (2009) [25] developed a controller to minimize pressure fluctuations in a wave-driven desalination system, and do consider the coupled dynamics of the WEC and PTO, but do not consider the trade-offs between smoothing and overall system performance. However, a more recent study by Sitterley et al. (2022) [26] shows that unsteady operation of SWRO is feasible and not as detrimental to membrane performance as typically feared. This finding opens the design space to investigate systems with less smoothing, although admittedly more research is needed to fully understand the techno-economic implications of unsteady SWRO operation.

When optimizing WEC systems, the optimal design of the device is highly dependent on the assumed wave conditions. Robertson et al. (2022) show that optimal WEC geometry is fundamentally dependant on the incident wave spectrum [27]. Due to the variability in incident wave conditions even at a single location, it is important to consider a variety of sea states when optimizing WEC systems. Many studies consider a variety of sea states when optimizing WEC systems, for example Stroofer et al. (2023) [9] use

a weighted average of the performance across multiple sea states as their objective function. This study takes a different approach by optimizing the design for various sea states individually, giving us a better understanding of the sensitivity of the optimal design to sea state.

This paper demonstrates the importance of wholistic modeling and co-design for wave-driven desalination systems. In Section 2, our formal MDO problem formulation is presented, defining our design space and an overview of the wholistic model. Section 3 goes into detail on the models for the various disciplines considered. Section 4 presents the results of our optimization studies which are then discussed in Section 5. Finally, conclusions and future work are presented in Section 6.

2. Problem Formulation

2.1. Multidisciplinary Design Optimization Framework

The optimization problem in this study is formally defined as:

$$\begin{aligned}
& \text{Minimize} && \text{LCOW}(\mathbf{x}, \mathbf{p}; \hat{u}) \\
& \text{by varying} && \mathbf{x} \\
& \text{subject to} && \mathbf{g} \leq 0 \\
& \text{while solving} && \mathbf{R}(\mathbf{x}, \mathbf{p}; \hat{u}) = 0 \\
& \text{for} && \hat{u}
\end{aligned}$$

LCOW (levelized cost of water) is the objective function (discussed further in Section 3.4). The design vector, $\mathbf{x}$, contains variables related to WEC, PTO, and SWRO plant, summarized in Table 1. The vector, $\mathbf{p}$, contains parameters related to environmental conditions and fixed design choices (shown in Appendix A). $\hat{u}$ represents the coupled variables solved for by $\mathbf{R}$, the governing equations of the various disciplinary modules: geometry, seawater desalination, hydrodynamics, system dynamics, and economics. The problem is subject to inequality constraints ($\mathbf{g}$) related to the piston motion and pressure which are enforced in the system dynamics module, Section 3.3.

Figure 3 presents a graphical representation of the MDO problem using an Extended Design Structure Matrix (xDSDM) [28]. The diagram uses gray line to illustrate the flow of information between the disciplinary modules and optimizer, while gray parallelograms indicate module inputs and outputs. The white parallelograms show optimizer level inputs and outputs.

2.2. Design Space

A summary of the design space is shown in Table 1. WEC variables consist of the width, w [m], thickness, t [m], and mass, m [kg], of the OSWEC.

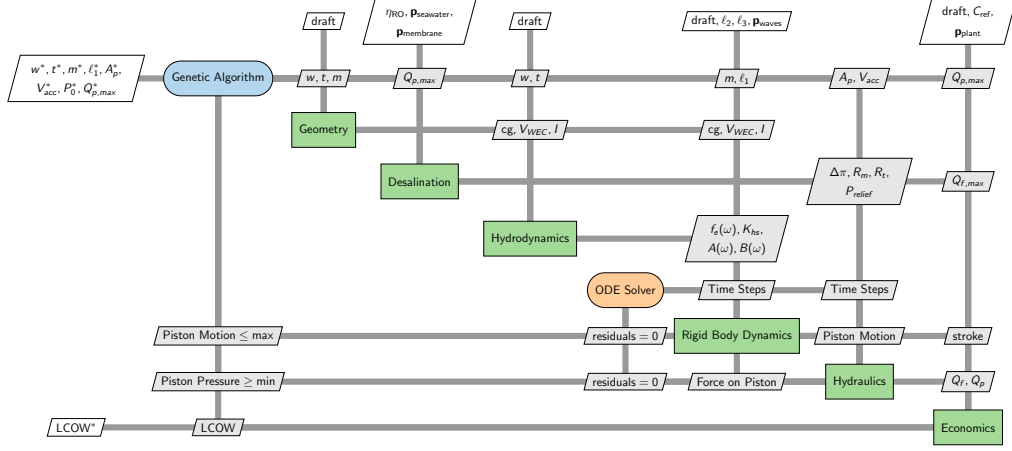


Figure 3: xDSM diagram of the wave-driven desalination system optimization problem. Note $\mathbf{p}_{seawater}$, $\mathbf{p}_{membrane}$, $\mathbf{p}_{waves}$, and $\mathbf{p}_{plant}$ represent vectors of parameters related to the membrane, seawater, sea state, and desalination plant architecture respectively.

Table 1: Design Variables

Variable	Nominal Value	Lower Bound	Upper Bound
w	18 m	10 m	30 m
t	2 m	1 m	5 m
m	127×10^3 kg	50×10^3 kg	500×10^3 kg
ℓ_1	2 m	0.1 m	draft
A_p	0.26 m ²	0.01 m ²	1 m ²
V_{acc}	4 m ³	0.01 m ³	6 m ³
P_0	3 MPa	1 MPa	7 MPa
$Q_{p,max}$	3150 m ³ /day	1000 m ³ /day	10000 m ³ /day

PTO variables include the OSWEC hinge to PTO joint length, ℓ_1 [m], piston area, A_p [m²], accumulator volume, V_{acc} [m³], and accumulator pre-charge pressure, P_0 [MPa]. Finally, the SWRO plant variable is the maximum plant production rate, $Q_{p,max}$ [m³/day]. Many of these variables can be visualized in Figure 4.

It is important to note that some design parameters are not included in the design space. WEC height and draft is set to fixed value, as are mechanism variables ℓ_2 and ℓ_3 . These were fixed to reduce design space dimensionality, as the other variables included provide sufficient control of the system dynamic characteristics (notably hydrodynamic stiffness, inertia, and excitation force, as well as the mechanical advantage of the mechanism).

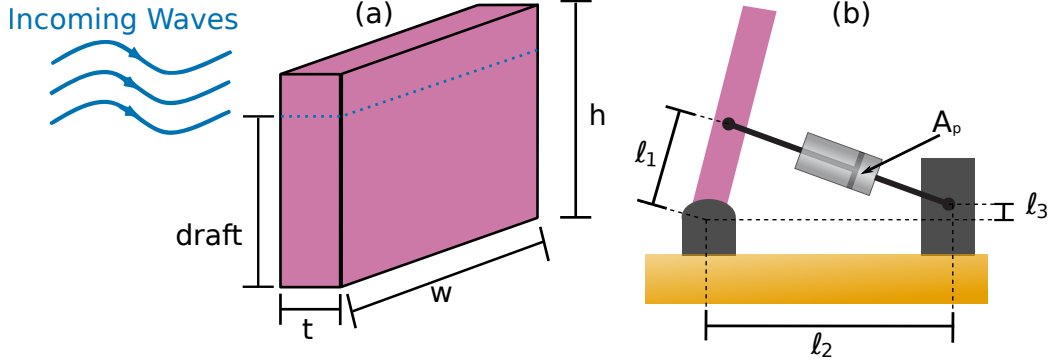


Figure 4: WEC (a) and mechanism (b) dimensions. Note that h represents the WEC height, which is a parameter rather than a design variable.

2.3. Parameters and Constraints

This study largely uses parameters similar to those used by Yu and Jenne (2018) [18]. Full tables of parameters can be found in Appendix A. Constraints are applied to device stroke length, maximum pressure in the hydraulic circuit, and minimum pressure in the piston cylinder. These are all enforced in the system dynamics module, Section 3.3.

2.4. Optimization Algorithm

Due to the absence of accessible gradients in the hydrodynamics and system dynamics modules, a genetic algorithm (GA) is utilized. A table of GA hyperparameters is provided in Appendix A.

2.5. Sequential Design Optimization Approach

MDO can be a computationally intensive task, which can limit its potential applications. In an effort to gauge the importance of MDO for this specific application a comparison is made between MDO and sequential design optimization (SDO) approaches. In SDO, we will optimize the design of each subsystem (WEC, PTO, and SWRO plant) sequentially. To optimize the WEC design, we optimize for levelized cost of WEC kinetic energy [\$/kWh] when disconnected from PTO. The only costs associated with this are the WEC costs.

To optimize the PTO and SWRO plant designs, we take two approaches. The first approach optimizes the PTO for levelized cost of flow [\$/m³] with the SWRO plant (membrane and pressure relief valve) replaced with a simple throttle valve. This results in optimizing the PTO to deliver the maximum feedflow into the SWRO plant at the lowest cost. This PTO is then used in the optimization of the SWRO plant, which optimizes the SWRO plant design for LCOW.

The second approach chooses a nominal PTO design from literature [18] and optimizes the SWRO plant design for LCOW using this fixed PTO design. This optimal SWRO plant design is then fixed while optimizing the PTO design for LCOW.

2.6. Sensitivity Analysis

The optimal design of the wave-driven desalination system is expected to change with different sea states. Since one of the goals of this study is to explore areas of potential design improvement through MDO, it is therefore important to see if the same design trends are present in a variety of sea states. To this end, optimizations are performed for 20 different sea states. The 20 sea states are selected using K-means clustering [29]. First, K-means clustering is performed on data from National Oceanographic and Atmospheric Administration (NOAA) National Data Buoy Center (NDBC) [30] for 5 different buoys. The 5 buoys locations chosen are: Guam (52200), Hilo HI (51206), Santa Monica Bay (46221), San Juan PR (41053), and Georges Bank (44011). 10 clusters are created for each buoy using hourly data from 2015-2024. From these 50 clusters, K-means clustering is performed again to create 20 final clusters used in the sensitivity study.

3. System Modeling

The xDSM in Figure 3 shows the 5 primary disciplinary modules: “Geometry”, which maps WEC design variables to geometric properties, “Desalination”, which calculates the system dynamic properties of the SWRO plant, “Hydrodynamics”, which determines various hydrodynamic coefficients, “System Dynamics”, which solves the coupled dynamics of the wave-driven desalination system, and “Economics”, which calculates the LCOW of the system. The “Geometry” module is straightforward and will not be discussed further. The remaining four modules are discussed in detail in the following subsections. This model framework was also presented and published by DeGoede and Haji (2025) [31].

3.1. Desalination Module

The desalination module determines key SWRO plant parameters for a single-stage plant. It considers the capacity of the plant in addition to membrane parameters and seawater composition. The governing equation for the reverse osmosis process is:

$$Q_p = A_w A_m (\Delta P - \Delta \pi), \quad (1)$$

where Q_p is the permeate flow rate [m^3/s], A_w is the water permeability coefficient [$\text{m}^3/\text{N}\cdot\text{s}$], A_m is the total membrane area [m^2], ΔP is the pressure difference across the membrane [Pa], and $\Delta\pi$ is the osmotic pressure difference across the membrane [Pa]. This osmotic pressure difference is calculated using [32]:

$$\pi = iCRT, \quad (2)$$

where i [-] is the number of ions produced per molecule of solute, C [mol/m^3] is the concentration of the solute, R [$\text{J}/\text{K}\cdot\text{mol}$] is the ideal gas constant, and T [K] is the temperature. For simplicity, $\Delta\pi$ is set as the osmotic pressure of the seawater minus the osmotic pressure of the target permeate concentration. A better model would consider the time variant concentrations on both sides of the membrane, but is outside the scope of this study.

The water permeability coefficient (A_w) is set to a constant value, matching the work by Yu and Jenne (2018) [18]. This constant value is dependent upon membrane selection, for this paper we chose to use the same membrane as Yu and Jenne (2018), the SW30HR-380 Dry from DuPont [33]. Membrane area A_m is set by calculating the amount of SW30HR-380 area is required to meet the SWRO plant capacity design variable, $Q_{p,max}$:

$$A_m = \frac{Q_{p,max}}{Q_0} A_0, \quad (3)$$

where Q_0 and A_0 are the single-unit production and membrane area for the SW30HR-380 membrane, found on the datasheet [33].

Next, for convenience a membrane resistance (R_m [$\text{MPa}\cdot\text{s}/\text{m}^3$]) is defined as:

$$\frac{1}{A_w A_m} \quad (4)$$

this is then used to determine the pressure relief valve set point, P_{relief} [MPa]:

$$P_{\text{relief}} = Q_{p,max} R_m + \Delta\pi \quad (5)$$

This equation ensures that if the flow exceeds the plant capacity, the pressure relief valve will open.

To force flow to pass through the membrane, resistance is required on the brine side. For this study, a simple throttle valve provides this resistance, but an ERU could also be used. The resistance of the throttle valve R_t [$\text{MPa}\cdot\text{s}/\text{m}^3$] is calculated as:

$$R_t = \frac{P_{\text{relief}}}{Q_{p,max}(\frac{1}{\eta_{RO}} - 1)} \quad (6)$$

where η_{RO} is the recommended recovery ratio of the SW30HR-380 membrane given the seawater composition, as determined using WAVE [34]. This equation ensures that when operating at full capacity, the recommended recovery ratio is used. When operating below capacity, the recovery ratio will be lower.

3.2. Hydrodynamics

Falnes and Kurniawan (2022) [35] define the oscillating WEC equation of motion as:

$$I\ddot{\xi} = f_e - f_r - f_b - f_v - f_f - f_u \quad (7)$$

where I [m or kg-m²] is the inertia of the WEC, ξ [m or rad] is the body motion, f_e [N or Nm] is the excitation force due to incident waves, f_r [N or Nm] is the radiation force due to the WEC's oscillations, f_b [N or Nm] is the hydrostatic force balancing buoyancy and weight, f_v [N or Nm] is the drag force arising from nonlinear viscous effects, f_f [N or Nm] captures the friction forces, and f_u [N or Nm] is the PTO force, representing the force applied by the WEC's power conversion system. The PTO force is modeled in the system dynamics module, Section 3.3. This study assumes linear potential flow theory and omits the drag and friction forces. This simplification is also found in Grasberger et al. (2024) [12].

The hydrostatic force accounts for the combination of buoyancy and weight and is defined as:

$$f_b = K_{hs}\xi, \quad (8)$$

where K_{hs} [N/m or Nm/rad] is the hydrostatic stiffness matrix. The radiation force accounts for the waves generated by the WEC's motion and is defined as:

$$f_r = A(\omega)\ddot{\xi} + B(\omega)\dot{\xi}, \quad (9)$$

where $A(\omega)$ [kg or kg-m²] is the added mass matrix and $B(\omega)$ [N-s/m or Nm-s/rad] is the radiation damping matrix. This formulation reduces the WEC hydrodynamics to 4 key coefficients: the hydrostatic stiffness, added mass, radiation damping, and excitation force.

These coefficients are determined using Capytaine [36], an open-source Boundary Element Method (BEM) solver. These solvers are well suited for wave-structure interaction analysis because they require meshing of the structure-fluid interface only. Computational fluid dynamics solvers require meshing of the entire fluid domain, this makes BEM solvers significantly less computationally expensive [37].

This study approximates the OSWEC as a rectangular prism, with one degree of freedom (pitch about the bottom hinge).

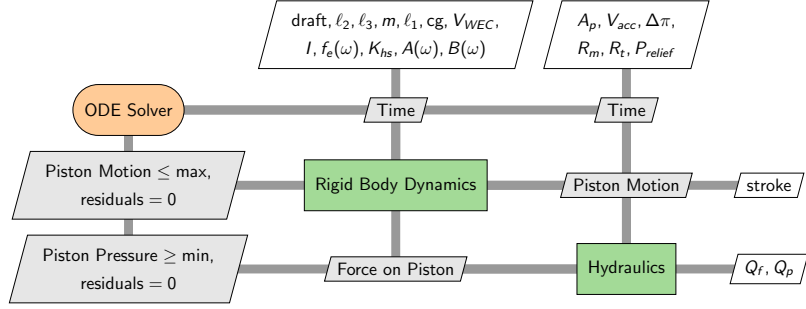


Figure 5: xDSM diagram of the system dynamics module.

3.3. System Dynamics

The system dynamics module solves the coupled dynamics of multiple disciplines simultaneously. Although it is presented as one discipline in the xDSM diagram in Figure 3, there are two coupled disciplines involved: rigid body dynamics and hydraulics. The interaction between these two subdisciplines is shown in the xDSM in Figure 5. These two disciplines are coupled using Simulink, with the piston motion from the rigid body dynamics module driving the hydraulic circuit in the hydraulics module, and the PTO force (f_u) from the hydraulics module feeding back into the rigid body dynamics module. MATLAB's ode4 solver, a fourth order Runge-Kutta solver, with a 0.1 s time step is used as the solver. The two subdisciplines are discussed in detail in the following sections.

3.3.1. Rigid Body Dynamics

The rigid body dynamics module considers the OSWEC dynamics as well as the dynamics of the mechanism connecting the WEC to the piston. The OSWEC dynamics are governed by Equation 7, with the various hydrodynamic coefficients determined through the BEM solver, as discussed in Section 3.2. WEC-Sim [38], an open-source software developed by the National Renewable Energy Laboratory (NREL) and Sandia National Laboratories, then uses these coefficients to solve the WEC dynamics in the time domain. The mechanism is modeled using the Simscape multibody toolbox, this models the kinematics and dynamics of the mechanism, relating the OSWEC pitch motion to the linear motion of the piston, as well as the force on the piston to the torque at the OSWEC hinge.

3.3.2. Hydraulics

The hydraulic circuit (Figure 2) is modeled using the isothermal liquid domain from the Simscape fluids toolbox. This domain is well suited for

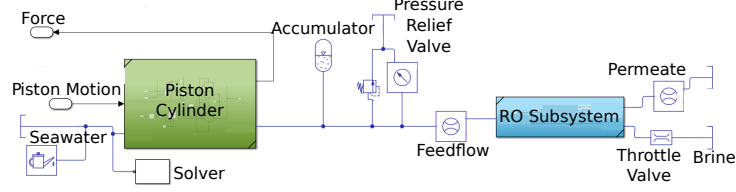


Figure 6: Simscape model of the hydraulic circuit.

modeling hydraulic systems. A high level view of the Simscape model is shown in Figure 6. This high level view shows the key components of the hydraulic circuit. The “Piston Cylinder” models the piston and directional valves. Flow out of the “Piston Cylinder” then passes through the accumulator, pressure relief valve, the “RO Subsystem” (which models the SWRO membrane), and finally the throttle valve on the brine side.

The “Piston Cylinder” is modeled similarly to the model used by Yu and Jenne (2018) [18]. The only notable difference is the use of the isothermal liquid domain rather than the hydraulics domain.

The “RO Subsystem” model is shown in Figure 7. The check valve prevents forward osmosis, and the membrane resistance block combined with the osmotic pressure block create a Simscape representation of Equation 1.

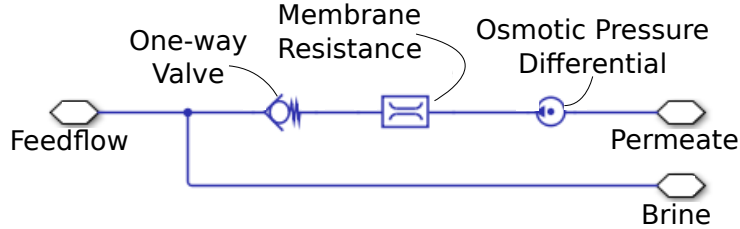


Figure 7: Simscape model of the RO membrane.

3.4. Economics

Our objective function is an adaptation of the levelized cost of energy (LCOE) [\$/kWh] function used by the U.S. Department of Energy [39]. Annual water production (AWP) [m³/yr] is substituted in for annual energy production [kWh/yr], creating this function for levelized cost of water (LCOW) [\$/m³]:

$$\text{LCOW} = \frac{(\text{FCR} \times \text{CAPEX}) + \text{OPEX}}{\text{AWP}}, \quad (10)$$

The fixed charge rate, FCR, is set to 10.8% as assumed in the U.S. Department of Energy report [39]. Annual water production (AWP) is calculated by scaling the results of the system dynamics simulation to a yearly basis.

The costs are then split into capital expenditures (CAPEX) [\$] and annual operating expenses (OPEX) [\$/yr].

To determine both CAPEX and OPEX, expenses are broken down into three cost models, WEC, PTO, and SWRO plant. These cost models are not indented to provide exact cost estimates. Rather, they are intended to quantify the relative costs of different designs to guide the optimization process. Although these cost models are not comprehensive, they serve as a tool to highlight the impact of MDO for wave-driven desalination systems. Each cost model is discussed in the following sections. Note that all costs are reported in 2025 USD.

3.4.1. WEC Cost Model

WEC cost modeling is challenging due to limited real world data, limiting accuracy. Despite this challenge, a practical cost model that captures key trends for design comparison is possible. In this study, we follow the approach used by Grasberger et al. (2024) [12] in their OSWEC CCD study. This model defines cost as a function of the float's surface area:

$$\text{Cost} = C_{1,ref} \left(\frac{A_x}{A_{ref}} \right) + C_{2,ref} \left(1 + \log \left(\frac{A_x}{A_{ref}} \right) \right), \quad (11)$$

where A_x [m²] is the float surface area and A_{ref} [m²] is the reference float surface area. $C_{1,ref}$ [\$] and $C_{2,ref}$ [\$] are the costs of the reference float associated with the two terms. Structural components are captured in the linear term while other expenses that are less dependent on size are captured in the logarithmic term. This cost model is dependent on accurate reference costs to make meaningful comparisons. For this study, reference costs are taken from NREL's Reference Model 5 (RM5) [40]. The costs from this reference model are tailored to this study by excluding the costs of irrelevant components, such as the electrical PTO system. Many transit and other overhead costs are also excluded as they are unlikely to scale with WEC size.

3.4.2. PTO Cost Model

The PTO cost covers the hydraulic cylinder and accumulator. The accumulator cost is estimated as a function of accumulator volume (V_{acc}):

$$\text{Cost} = 1.621 \times 10^5 V_{acc}^{0.986}, \quad (12)$$

where the constants are tuned to approximate supplier quotes [41].

The hydraulic cylinder cost is a function of the required amount of steel:

$$\text{Cost} = C_{316} \rho_{316} V_{316} (1 + L), \quad (13)$$

where C_{316} [\$/lb] is the unit cost of 316 stainless steel, ρ_{316} [lb/in³] is the density of 316 stainless steel, V_{316} [in³] is the volume of 316 stainless steel required, and L [-] is a labor multiplier. The volume required is the sum of the required volumes for the various parts of the cylinder: the cylinder barrel (V_{cylinder}) [in³], the end caps (V_{cap}) [in³], the piston head (V_{piston}) [in³], and the piston rod (V_{rod}) [in³].

$$V_{316} = V_{\text{cylinder}} + 2V_{\text{cap}} + V_{\text{piston}} + V_{\text{rod}}, \quad (14)$$

The thicknesses of the cylinder barrel and end caps are calculated according to the ASME Boiler and Pressure Vessel Code [42]. The piston head uses the same thickness as the end caps, while the piston rod diameter is set to avoid buckling and tensile/compressive failure. The required stroke length, which is needed for the cylinder volume calculation, is determined from the system dynamics simulation.

3.4.3. SWRO Cost Model

SWRO plant cost models on the scale of this paper (producing thousands of m³/day) are scarce and poorly documented. Most well documented cost models focus on large-scale plants (hundreds of thousands of m³/day) [43, 44, 45, 46], and are therefore not suitable for this study. Studies that examine plants of comparable size typically lack transparent cost models [6, 47, 48].

This study aims to use a transparent cost model for small-scale SWRO plants. Voutchkov's *Desalination Project Cost Estimating and Management* [49] provides cost curves for CAPEX (section 4) and OPEX (section 5) for a wide range of plant sizes. Using different curves can be used to model different plant architectures, giving this model additional flexibility. In this study, each curve is fitted to a power function, which prevents unrealistic behavior at the lower end of the scale.

4. Results

5. Discussion

6. Conclusion and Future Work

References

- [1] J. Eliasson, The rising pressure of global water shortages, *Nature* 517 (6) (Jan. 2015). doi:<https://doi.org/10.1038/517006a>.

- [2] Z. Li, A. Siddiqi, L. D. Anadon, V. Narayanamurti, Towards sustainability in water-energy nexus: Ocean energy for seawater desalination, *Renewable and Sustainable Energy Reviews* 82 (2018) 3833–3847. doi:10.1016/j.rser.2017.10.087.
- [3] U. D. of Energy, Powering the blue economy: Exploring opportunities for marine renewable energy in maritime markets (4 2019).
- [4] J. Gillis, California drought is made worse by global warming, scientists say, *New York Times* (Aug. 2015).
URL <http://web-static-aws.seas.harvard.edu/climate/eli/Courses/global-change-debates/Sources/California-droughts/CaliforniaDroughtIsMadeWorsebyGlobalWarming,ScientistsSay-TheNewYorkTimes.pdf>
- [5] L. Garcia-Rodriguez, V. Romero-Ternero, C. Gomez-Camacho, Economic analysis of wind-powered desalination, *Desalination* 137 (2001) 259–265. doi:[https://doi.org/10.1016/S0011-9164\(01\)00235-1](https://doi.org/10.1016/S0011-9164(01)00235-1).
- [6] M. R. Elkadeem, K. M. Kotb, S. W. Sharshir, M. A. Hamada, A. E. Kabeel, I. K. Gabr, M. A. Hassan, M. Y. Worku, M. A. Abido, Z. Ullah, H. M. Hasanien, F. F. Selim, Optimize and analyze a large-scale grid-tied solar pv-powered swro system for sustainable water-energy nexus, *Desalination* 579 (2024) 117440. doi:10.1016/j.desal.2024.117440.
- [7] P. Davies, Wave-powered desalination: resource assessment and review of technology, *Desalination* 186 (1–3) (2005) 97–109. doi:10.1016/j.desal.2005.03.093.
- [8] J. Sobieszczanski-Sobieski, Multidisciplinary design optimization: An emerging new engineering discipline, *Solid Mechanics and Its Applications* (1995) 483–496doi:10.1007/978-94-011-0453-1_14.
- [9] C. A. M. Ströfer, D. T. Gaebele, R. G. Coe, G. Bacelli, Control co-design of power take-off systems for wave energy converters using wecopttool, *IEEE Transactions on Sustainable Energy* 14 (4) (2023) 2157–2167. doi:10.1109/tste.2023.3272868.
- [10] Y. Peña-Sanchez, D. García-Violini, J. V. Ringwood, Control co-design of power take-off parameters for wave energy systems, *IFAC-PapersOnLine* 55 (27) (2022) 311–316. doi:10.1016/j.ifacol.2022.10.531.

- [11] M. Rosati, J. V. Ringwood, Control co-design of power take-off and bypass valve for owc-based wave energy conversion systems, *Renewable Energy* 219 (2023) 119523. doi:10.1016/j.renene.2023.119523.
- [12] J. Grasberger, L. Yang, G. Bacelli, L. Zuo, Control co-design and optimization of oscillating-surge wave energy converter, *Renewable Energy* 225 (2024) 120234. doi:10.1016/j.renene.2024.120234.
- [13] P. B. Garcia-Rosa, J. V. Ringwood, On the sensitivity of optimal wave energy device geometry to the energy maximizing control system, *IEEE Transactions on Sustainable Energy* 7 (1) (2016) 419–426. doi:10.1109/TSTE.2015.2423551.
- [14] R. G. Coe, G. Bacelli, D. Forbush, A practical approach to wave energy modeling and control, *Renewable and Sustainable Energy Reviews* 142 (represent) (2021) 110791. doi:10.1016/j.rser.2021.110791.
- [15] R. G. Coe, G. Bacelli, D. Gaebale, A. Keow, D. Forbush, Co-design of a wave energy converter through bi-conjugate impedance matching, *Mechatronics* 111 (2025) 103395. doi:<https://doi.org/10.1016/j.mechatronics.2025.103395>.
URL <https://www.sciencedirect.com/science/article/pii/S0957415825001047>
- [16] S. Ladan, K. Wu, Nonlinear modeling and harmonic recycling of millimeter-wave rectifier circuit, *IEEE Transactions on Microwave Theory and Techniques* 63 (3) (2015) 937–944. doi:10.1109/TMTT.2015.2396043.
- [17] D. J. Yi-Hsiang Yu, Analysis of a wave-powered, reverse-osmosis system and its economic availability in the united states, in: *Proceedings of the ASME 2017 36th International Conference on Ocean, Offshore and Arctic Engineering OMAE2017*, 2017. doi:<https://doi.org/10.1115/OMAE2017-62136>.
- [18] Y.-H. Yu, D. Jenne, Numerical modeling and dynamic analysis of a wave-powered reverse-osmosis system, *Journal of Marine Science and Engineering* 6 (4) (2018) 132. doi:10.3390/jmse6040132.
- [19] K. M. Brodersen, E. A. Bywater, A. M. Lanter, H. H. Schennum, K. N. Furia, M. K. Sheth, N. S. Kiefer, B. K. Cafferty, A. K. Rao, J. M. Garcia, D. M. Warsinger, Direct-drive ocean wave-powered batch reverse osmosis, *Desalination* 523 (2022) 115393. doi:10.1016/j.desal.2021.115393.

- [20] R. Suchithra, T. K. Das, K. Rajagopalan, A. Chaudhuri, N. Ulm, M. Prabu, A. Samad, P. Cross, Numerical modelling and design of a small-scale wave-powered desalination system, *Ocean Engineering* 256 (2022) 111419. doi:10.1016/j.oceaneng.2022.111419.
- [21] J. W. Simmons, J. D. Van de Ven, A comparison of power take-off architectures for wave-powered reverse osmosis desalination of seawater with co-production of electricity, *Energies* 16 (21) (2023). doi:10.3390/en16217381.
URL <https://www.mdpi.com/1996-1073/16/21/7381>
- [22] J. Mi, X. Wu, J. Capper, X. Li, A. Shalaby, R. Wang, S. Lin, M. Hajj, L. Zuo, Experimental investigation of a reverse osmosis desalination system directly powered by wave energy, *Applied Energy* 343 (2023) 121194. doi:10.1016/j.apenergy.2023.121194.
- [23] W. Lai, Q. Ma, H. Lu, S. Weng, J. Fan, H. Fang, Effects of wind intermittence and fluctuation on reverse osmosis desalination process and solution strategies, *Desalination* 395 (2016) 17–27. doi:10.1016/j.desal.2016.05.019.
- [24] B. J. Hopkins, N. Padhye, A. Greenlee, J. Torres, L. Thomas, D. M. Ljubicic, M. P. Kassner, A. H. Slocum, Damping pressure pulsations in a wave-powered desalination system, *Journal of Energy Resources Technology* 136 (2) (Apr. 2014). doi:10.1115/1.4026635.
- [25] G. Bacelli, J.-C. Gilloteaux, J. Ringwood, A predictive controller for a heaving buoy producing potable water, in: *European Control Conference*, 2009.
- [26] K. A. Sitterley, T. J. Cath, D. S. Jenne, Y.-H. Yu, T. Y. Cath, Performance of reverse osmosis membrane with large feed pressure fluctuations from a wave-driven desalination system, *Desalination* 527 (2022) 115546. doi:10.1016/j.desal.2022.115546.
- [27] B. Robertson, G. Dunkle, T. Mundon, L. Kilcher, Wave resource spatial and temporal variability dependence on wec size, *International Marine Energy Journal* 5 (1) (2022) 113–121. doi:10.36688/imej.5.113-121.
URL <https://marineenergyjournal.org/imej/article/view/104>
- [28] A. B. Lambe, J. R. R. A. Martins, Extensions to the design structure matrix for the description of multidisciplinary design, analysis, and optimization processes, *Structural and Multidisciplinary Optimization*

- 46 (2) (2012) 273–284. doi:10.1007/s00158-012-0763-y.
URL <https://doi.org/10.1007/s00158-012-0763-y>
- [29] D. Arthur, S. Vassilvitskii, k-means++: the advantages of careful seeding, in: Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms, SODA '07, Society for Industrial and Applied Mathematics, USA, 2007, p. 1027–1035.
- [30] U.S. National Data Buoy Center, National data buoy center – real-time oceanographic and meteorological data, <https://www.ndbc.noaa.gov/> (1971).
- [31] N. DeGoede, M. Haji, A multidisciplinary design optimization framework for wave-driven desalination systems, Vol. Volume 3B: 51st Design Automation Conference (DAC) of International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, 2025, p. V03BT03A010. arXiv:<https://asmedigitalcollection.asme.org/IDETC-CIE/proceedings-pdf/IDETC-CIE2025/89237/V03BT03A010/7557602/v03bt03a010-detc2025-168312.pdf>, doi:10.1115/DETC2025-168312.
URL <https://doi.org/10.1115/DETC2025-168312>
- [32] J. D. Seader, E. J. Henley, D. K. Roper, SEPARATION PROCESS PRINCIPLES: Chemical and Biochemical Operations, 3rd Edition, Wiley, 2011.
- [33] DuPont, Filmtec™ sw30hr-380 element product data sheet, Datasheet (Oct. 2024).
- [34] DuPont, Wave design software.
- [35] J. Falnes, A. Kurniawan, Ocean Waves and Oscillating Systems: Linear Interactions Including Wave-Energy Extraction, 2nd Edition, Cambridge University Press, 2020. doi:10.1017/9781108674812.
- [36] M. Ancellin, F. Dias, Capytaine: a Python-based linear potential flow solver, Journal of Open Source Software 4 (36) (2019) 1341. doi:10.21105/joss.01341.
URL <https://doi.org/10.21105%2Fjoss.01341>
- [37] J. Newman, C.-H. Lee, Boundary-element methods in offshore structure analysis, Journal of Offshore Mechanics and Arctic Engineering 124 (2002) 81. doi:10.1115/1.1464561.

- [38] Wec-sim (wave energy converter simulator).
URL <https://wec-sim.github.io/WEC-Sim/master/index.html>
- [39] A. LaBonte, P. O'Connor, C. Fitzpatrick, K. Hallett, Y. Li, Standardized cost and performance reporting for marine and hydrokinetic technologies, in: 1st Marine Energy Technology Symposium, Washington DC, 2013.
- [40] D. J. Y.-H. Yu, S. G. R. Thresher, A. Copping, L. Hanna, Reference model 5 (rm5): Oscillating surge wave energy converter, Tech. Rep. Report (Jan. 2015).
- [41] ReasonTek, Low pressure accumulator quote (03 2025).
- [42] American Society of Mechanical Engineers, Boiler and Pressure Vessel Code, Section VIII, Division 1: Rules for Construction of Pressure Vessels, ASME, New York, 2015, aSME BPVC-VIII-1-2015.
- [43] A. H. Slocum, M. N. Haji, A. Z. Trimble, M. Ferrara, S. J. Ghaemsaidi, Integrated pumped hydro reverse osmosis systems, Sustainable Energy Technologies and Assessments 18 (2016) 80–99. doi:10.1016/j.seta.2016.09.003.
- [44] M. W. Haefner, M. N. Haji, Integrated pumped hydro reverse osmosis system optimization featuring surrogate model development in reverse osmosis modeling, Applied Energy 352 (2023) 121812. doi:10.1016/j.apenergy.2023.121812.
- [45] R. Huehmer, J. Gomez, J. M. Curl, K. Moore, Cost modeling of desalination systems, in: International Desalination Association World Congress, International Desalination Association World Congress, International Desalination Association World Congress, 2011.
URL https://www.researchgate.net/publication/299408931_COST_MODELING_OF_DESALINATION_SYSTEMS
- [46] M. K. Wittholz, B. K. O'Neill, C. B. Colby, D. Lewis, Estimating the cost of desalination plants using a cost database, Desalination 229 (1–3) (2008) 10–20. doi:10.1016/j.desal.2007.07.023.
- [47] M. Gökçek, O. B. Gökçek, Technical and economic evaluation of freshwater production from a wind-powered small-scale seawater reverse osmosis system (wp-swro), Desalination 381 (site.) (2016) 47–57. doi:10.1016/j.desal.2015.12.004.

- [48] I. A. E. Agency, Deep 5 user manual, 2013.
- [49] N. Voutchkov, Desalination project cost estimating and management, CRC Press, 2019.
- [50] Shane, Comprehensive guide to stainless steel pricing and grades, accessed: 2025-05-14 (March 2025).
URL <https://shop.machinemfg.com/comprehensive-guide-to-stainless-steel-pricing/>

Appendix A. TABLES OF PARAMETERS

Table A.2: GENETIC ALGORITHM HYPERPARAMETERS

Parameter	Value	Units
population size	256	-
number of generations	300	-
mutation rate	0.02	-
crossover rate	0.8	-
number of elites kept	3	-
tournament size	4	-
bits per variable	8	-

Table A.3: GENERAL PARAMETERS

Parameter	Value	Units
gravitational acceleration	9.81	m/s ²
ocean density	1025	kg/m ³
distance to shore	500	m
ocean temperature	298.15	K
water depth	12	m
wave direction	0 [18]	degrees
wave spectrum	Pierson–Moskowitz [18]	-
significant wave height	2.64 [18]	m
peak period	9.86 [18]	s
Fixed Charge Rate	10.8 [39]	%

Table A.4: SWRO PARAMETERS

Parameter	Value	Units
feedflow total dissolved solids	35946 [18]	mg/L
permeate total dissolved solids	150 [18]	mg/L
salt molar weight	58.44	g/mol
water permeability coefficient	2.57×10^{-12} [18]	m ³ /N-s
recovery ratio	44.2 [34]	%
single SW30HR-380 area	35 [33]	m ²
single SW30HR-380 flow rate	24.6 [33]	m ³ /day

Table A.5: WEC PARAMETERS

Parameter	Value	Units
draft	9	m
cg draft factor	-0.7778	-
unit inertia	14.57	m ²
RM5 surface area	1214 [40]	m ²
RM5 flap cost	3364648.63 [40]	\$
RM5 base cost	1706415.27 [40]	\$
RM5 bearings cost	17420.34 [40]	\$
RM5 mooring cost	997819.2 [40]	\$
RM5 monitoring cost	616480.27 [40]	\$/yr
RM5 marine operations cost	101387.23 [40]	\$/yr
RM5 shore operations cost	347280.29 [40]	\$/yr
RM5 parts cost	86237.2 [40]	\$/yr
RM5 consumables cost	17480.19 [40]	\$/yr
RM5 insurance rate	2 [40]	%

Table A.6: PTO PARAMETERS

Parameter	Value	Units
ℓ_2	4.7	m
ℓ_3	0	m
max piston stroke	20	m
SS316 cost	2.00 [50]	\$/in ³
SS316 density	0.29 [50]	lb/in ³
SS316 yield strength	206 [50]	MPa
SS316 Young's modulus	164 [50]	GPa
cylinder factor of safety	6	-
rod factor of safety	1.5	-
labor factor	0.7	-
end cap attachment factor	0.3 [42]	-
cylinder joint efficiency	0.8 [42]	-

Table A.7: SOLVER PARAMETERS

Parameter	Value	Units
BEM frequencies	0.2, 0.34, 0.48, ..., 3	rad/s
SysDyn time step	0.1	s
SysDyn sim time	300	s